Prabhavit Gupta

AI Engineer | Agentic Pipelines, LLM Orchestration, Evals & Production Systems

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Executive Summary

- **Agentic Pipelines & Commerce Intelligence:** AI Engineer with 3 years commercial experience architecting and shipping production agentic workflows, autonomous conversational pipelines, and high-throughput vector retrieval systems. Experienced taking 0-to-1 agent systems into production with tight latency and evaluation constraints.
- **LLM Orchestration, Tooling & Hybrid Recommendation:** Deep practical expertise with native Anthropic/OpenAI SDKs, LangGraph, Model Context Protocol (MCP), and hybrid retrieval (vector similarity + BM25/fuzzy keyword search) across PostgreSQL (pgvector), Redis, and Qdrant.
- **Production Evals, Observability & Rigor:** Proven track record designing automated evaluation loops (RAGAS, DeepEval, prompt regression suites) and real-time distributed tracing with Langfuse. Systematic approach to tracking latency, token economics, failure cascades, and accuracy drift across conversational datasets.

Technical Skills & Core Competencies

- **AI & Agentic Systems:** Agentic Frameworks (LangGraph, LangChain), Native OpenAI/Anthropic SDKs, Autonomous Agent Loops, Tool Use & Model Context Protocol (MCP), Context Pruning, Guardrails.
- **LLM Observability & Evals:** Langfuse Distributed Tracing, RAGAS Benchmarking, DeepEval, Synthetic Evaluation Datasets, Prompt Regression Testing, PostHog Event Analytics.
- **Vector Search & Recommenders:** PostgreSQL (pgvector, HNSW indexing), Hybrid Scoring (Semantic + Lexical BM25/pg_trgm), Redis Semantic Caching, Qdrant, Data Ingestion Pipelines.
- **Engineering & Infrastructure:** TypeScript / Node.js, Python (AsyncIO, FastAPI, Pydantic), Docker, GCP, PostgreSQL, Redis, REST & WebSocket APIs, GitHub Actions CI/CD.
- **Development Philosophy:** AI-Assisted Engineering (Claude Code, Cursor), Rapid 0-to-1 Prototyping, Production Scalability, Extreme Ownership.

Professional Experience

AI Engineer & Solutions Consultant • *Independent / AI Product Engineering Consultancy*
2024 – Present • Venice, IT (Remote Europe)

- Architected and deployed production multi-agent workflows and real-time LLM pipelines for European tech clients, serving real-world conversational and data extraction use cases.
- Designed end-to-end agentic pipelines with autonomous tool calling, structured JSON output validation, and state persistence using LangGraph and Model Context Protocol (MCP).
- Implemented hybrid recommendation and retrieval engines pairing dense vector embeddings (pgvector HNSW) with lexical search, optimizing sub-second query latency under high request loads.
- Instrumenting comprehensive distributed observability via Langfuse, capturing end-to-end agent decision traces, token consumption, and fallback behavior across millions of events.
- Built automated evaluation and regression harnesses using RAGAS and custom LLM-as-a-judge pipelines, establishing quantitative benchmarks for output relevance, groundedness, and hallucination prevention.
- Integrated automated data enrichment and semantic classification loops, keeping internal product catalogs and knowledge bases continuously synchronized.

Full-Stack AI Software Engineer • *Tech Automation Systems*
2023 – 2024 • Milan / Remote, IT

- Engineered high-throughput asynchronous services in TypeScript and Python for conversational intent extraction and entity categorization.
- Implemented semantic caching mechanisms in Redis and PostgreSQL pgvector, cutting redundant API calls by 35% and stabilizing response latency under peak traffic.
- Deployed microservices via Docker and automated CI/CD pipelines with GitHub Actions, enforcing strict unit testing and schema validation gates.
- Collaborated with product leads to translate user interaction signals into measurable algorithmic iteration loops and structured feature roadmaps.

Key Engineering Projects

- **Autonomous Agentic Commerce Pipeline (TypeScript + LangGraph + pgvector):** Built an autonomous research and recommendation agent that extracts catalog attributes, queries vector stores with hybrid scoring, and generates structured comparison cards with Langfuse telemetry.
- **Continuous LLM Evaluation & Regression Suite (Python + RAGAS + DeepEval):** Designed an automated CI/CD evaluation gate that runs prompt regression tests against synthetic customer interaction benchmarks,

detecting latency regressions and semantic drift prior to deployment.

Education, Languages & Legal Status

- **Bachelor of Arts in Philosophy & Economics** – Ca' Foscari University of Venice (2018 – 2022)
  Rigorous foundation in formal logic, statistical decision theory, computational models, and analytical philosophy.
- **Languages:** English (C1 Full Professional / Fluent), Italian (Working proficiency), French (Basic).
- **Work Authorization & Legal Status:** EU Long-Term Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy) – Full legal authorization to work in France / EU without visa sponsorship; available immediately.